ASEEM BARANWAL

EDUCATION

Ph.D. in Computer Science, University of Waterloo — Sep 2020 – Present
Supervised by Aukosh Jagannath and Kimon Fountoulakis

M.Math. (thesis) in Computer Science, University of Waterloo — Sep 2018 – Apr 2020
Supervised by Jeffrey Shallit

B.Tech. in Computer Science, Indian Institute of Technology Jodhpur — Jul 2012 – May 2016

EXPERIENCE

- **Research Intern, Microsoft Research** — Jun 2023 – Sep 2023
  Worked on graph learning models for polymorphic malware detection.
- **Research Intern, Google Research** — Oct 2022 – Dec 2022
  Worked with the Graph mining group on scalable GNNs for sparse graphs.
- **Software Engineer, Microsoft Corporation** — Jun 2016 – Jul 2018
  Worked with the Azure compute group on distributed systems and anomaly detection.

PUBLICATIONS & MANUSCRIPTS

- **Analysis of Corrected Graph Convolutions**. R. Wang, A. Baranwal, K. Fountoulakis. Neural Information Processing Systems (NeurIPS), 2024.
- **Optimality of Message-passing Architectures for Sparse Graphs**. A. Baranwal, K. Fountoulakis, A. Jagannath. Neural Information Processing Systems (NeurIPS), 2023.
- **Graph Attention Retrospective**. K. Fountoulakis, A. Levi, S. Yang, A. Baranwal, A. Jagannath. Journal of Machine Learning Research (JMLR) vol. 24:246 pp. 1–52, 2023.
- **Effects of Graph Convolutions in Multi-layer Networks**. A. Baranwal, K. Fountoulakis, A. Jagannath. International Conference on Learning Representations (ICLR), 2023. *(Spotlight)*
- **Graph Convolution for Semi-supervised Classification: Improved Linear Separability and Out-of-distribution Generalization**. A. Baranwal, K. Fountoulakis, A. Jagannath. International Conference on Machine Learning (ICML), 2021. *(Spotlight)*
- **Antisquares and Critical Exponents**. A. Baranwal, J. Currie, L. Mol, P. Ochem, N. Rampersad, J. Shallit. Discrete Mathematics & Theoretical Computer Science vol. 25:2, 2023.
- **Ostrowski-automatic Sequences: Theory and Applications**. A. Baranwal, L. Schaeffer, J. Shallit. Theoretical Computer Science 858, pp. 122–142, 2021.
- **Decision Algorithms for Ostrowski-automatic Sequences.** A. Baranwal. MMath Thesis, University of Waterloo, School of Computer Science, 2020.
- **Repetitions in Infinite Palindrome-rich Words**. A. Baranwal, J. Shallit. In Combinatorics on Words. WORDS 2019. LNCS vol. 11682, Springer, pp. 93–105, 2019.
- **Critical Exponent of Balanced Words via the Pell Number System**. A. Baranwal, J. Shallit. In Combinatorics on Words. LNCS vol. 11682, Springer, pp. 80–92, 2019.

AWARDS AND ACHIEVEMENTS

- Cheriton Graduate Scholarship, University of Waterloo (10,000 CAD) 2024
- President's Graduate Scholarship, University of Waterloo (10,000 CAD) 2023–2024
- Ontario Graduate Scholarship, Province of Ontario and UWaterloo (15,000 CAD) 2023–2024
- Best paper award, GroundedML workshop at ICLR 2022
- Top reviewer award, Learning on Graphs conference 2022
- Graduate Excellence Award, University of Waterloo 2020–2022
- Graduate Research Travel Grant, GSPA, University of Waterloo 2019
- Microsoft Engineering Star Award – Changing Status Quo 2017
- Team ranked 14 at ACM ICPC Asia regionals 2015

TALKS

- Locally Optimal Message-passing on Feature-decorated Sparse Graphs, BIRS, Banff May 2024
- Optimality of Message-passing Architectures for Sparse Graphs, NeurIPS, New Orleans Dec 2023
- Polymorphic Malware Classification using Graph Neural Networks, Microsoft Research Aug 2023
- Effects of Graph Convolutions in Multi-layer Networks, ICLR (virtual talk) Apr 2023
- Graph Convolutions for semi-supervised learning, ICML (virtual talk) Jul 2021
- Repetitions in infinite palindrome-rich words, Loughborough University, England Sep 2019
- Anomaly detection and monitoring of metrics in Azure services, Microsoft Corporation Jan 2017

TEACHING

- Instructional Apprentice, CS 341 - Algorithms 2019–2024
- Teaching Assistant, CS 135 - Designing Functional Programs 2018–2019

MISCELLANEOUS

- Serving as a reviewer for NeurIPS, ICLR, LoG, UAI 2021–Now
- Received talent bursaries from the Alberta Machine Intelligence Institute 2023, 2024
- Helped underprivileged kids with education at SOS Children's Villages of India 2016–2018
- Problem setter for ACM ICPC preliminary contests in India 2015–2016